Context Buys Precision, Not Recall: A Transformer Successor to a Lexical Baseline for Propaganda Detection

Working Paper II in Computational Political Anthropology

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Politomorphism Engine — Defensive Political Anthropology Layer

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Code & data: github.com/politomorphism/politomorphism-defense

Abstract

A prior working paper established a transparent TF-IDF and logistic-regression baseline for propaganda detection on the English partition of the PropXplain corpus, reaching an F1 of 0.506 on the propagandistic class, and argued from its errors that fine political manipulation is carried by discourse structure rather than by vocabulary. This paper tests the direct consequence of that argument: that a context-aware transformer should recover the structural manipulation the lexical model misses — that is, that it should improve recall on the propagandistic class. I fine-tune DistilBERT on the same data, with the same class-balancing and the same evaluation protocol. The transformer improves the propagandistic-class F1 modestly, from 0.506 to 0.549, but the improvement is located almost entirely in precision, which rises from 0.465 to 0.555 as false positives fall from 160 to 109. Recall is effectively flat (0.556 to 0.544) and false negatives are essentially unchanged (111 to 114). The central hypothesis is therefore not supported: at this scale, contextual representation makes the detector more discriminating — it stops over-flagging — but it does not recover the subtle manipulation it was expected to catch. I report this as a single-run result whose magnitude is within plausible noise, examine why recall may be resistant, and argue that the recovery of fine, structural manipulation remains the open problem of the defensive audit layer of the Politomorphism Engine.

Keywords: propaganda detection; transformers; DistilBERT; precision and recall; class imbalance; computational political anthropology; defensive auditing; Politomorphism

1. Introduction

A prior working paper in this programme established a deliberately transparent baseline for the detection of political manipulation: a TF-IDF and logistic-regression classifier trained on the English partition of the PropXplain corpus, which reached an F1 of 0.506 on the propagandistic class. That paper's central argument came not from its score but from its errors. A lexical model, built from word and bigram frequencies, systematically failed on passages whose manipulation lay not in charged vocabulary but in structure — in presupposition smuggled into a subordinate clause, in rhetorical equivalence between unlike things, in what a sentence asks the reader to accept without stating. From this the paper drew a claim about the phenomenon: that fine political manipulation is frequently contextual rather than lexical. The present paper, the second in the programme, puts that claim to a direct test. Its code and data are maintained in the same public repository at github.com/politomorphism/politomorphism-defense.

2. Hypothesis

If fine manipulation is carried by discourse structure rather than vocabulary, then a model capable of representing context — the dependency between a clause and its presupposition, the relation between a framing and the thing it frames — should recover the manipulation the lexical baseline misses. Operationally, this predicts a specific movement in the metrics: the contextual model should raise recall on the propagandistic class, because the cases the baseline missed are precisely the structurally manipulative ones. Precision might rise or fall; the decisive prediction concerns recall. This paper tests that prediction by replacing the lexical features with a fine-tuned transformer while holding everything else — corpus, split, class-balancing, and evaluation protocol — constant.

3. Data and Method

The data are unchanged from the baseline: the English configuration of PropXplain, with 4,472 training paragraphs and 922 held-out test paragraphs, labelled propagandistic or non-propagandistic, with the same imbalance (non-propagandistic paragraphs are roughly 73% of the test set). The model is DistilBERT (distilbert-base-uncased), fine-tuned for sequence classification. To hold the treatment of imbalance identical in spirit to the baseline's balanced class weights, the training objective is a class-weighted cross-entropy whose weights are the inverse class frequencies. Training runs for three epochs with a learning rate of 2e-5, a maximum sequence length of 256 tokens, and a fixed random seed; inputs longer than 256 tokens are truncated. Evaluation is performed once on the untouched test partition, using exactly the metrics of the baseline paper, with the propagandistic class as the positive class. Because training was performed on CPU, the configuration is modest by design; the single-run, fixed-seed nature of the result is a limitation returned to in Section 6.

4. Results

Table 1 reports the transformer's per-class performance on the held-out test partition. The non-propagandistic figures and the accuracy are computed directly from the confusion matrix and are therefore exact.

Class	Precision	Recall	F1	Support
non-propagandistic	0.832	0.838	0.835	672
propagandistic	0.555	0.544	0.549	250
Accuracy			0.758	922
Macro avg.	0.693	0.691	0.692	922
Weighted avg.	0.757	0.758	0.757	922

Table 1. Per-class performance of the fine-tuned DistilBERT detector on the PropXplain English test partition (n = 922).

Table 2 places the two models side by side on the propagandistic class, which is the defensively relevant one, and adds the raw error counts that make the comparison concrete.

Measure (propagandistic class)	Baseline (TF-IDF)	Transformer (DistilBERT)	Δ
Precision	0.465	0.555	+0.090
Recall	0.556	0.544	−0.012
F1	0.506	0.549	+0.043
False positives	160	109	−51

Measure (propagandistic class)	Baseline (TF-IDF)	Transformer (DistilBERT)	Δ
False negatives	111	114	+3
Accuracy (overall)	0.706	0.758	+0.052

Table 2. Baseline (TF-IDF + logistic regression) versus transformer (DistilBERT) on the propagandistic class. Green marks improvement, red marks regression. Both are single evaluation runs.

The transformer improves the propagandistic-class F1 from 0.506 to 0.549. The whole of that gain, however, is a gain in precision: precision rises from 0.465 to 0.555 as the model's false positives fall from 160 to 109. Recall moves in the opposite, unhelpful direction, from 0.556 to 0.544, and the number of genuinely propagandistic paragraphs the model misses is essentially unchanged — 114 false negatives, against the baseline's 111. Overall accuracy rises from 0.706 to 0.758, but, as argued in the first paper, accuracy is not the defensively meaningful quantity here.

5. Analysis

The hypothesis is not supported in its central claim. The prediction was an improvement in recall, on the reasoning that the baseline's misses were structurally manipulative cases a contextual model would recover. Recall did not improve; it fell slightly, and the false-negative count barely moved. What the transformer learned, instead, was to stop over-flagging: it reclassified some fifty previously false-positive paragraphs correctly as non-propagandistic, at the cost of three true positives. In defensive terms, the contextual model became more discriminating about what is not manipulation, not more sensitive to the manipulation that is hard to see. The subtle, structural cases that motivated the move to a transformer remain, for the most part, undetected.

This is a more interesting result than a clean confirmation would have been, because it separates two things the first paper's argument had tacitly joined. That fine manipulation is structural may well be true as a description of the phenomenon; but it does not follow that a modest contextual model, trained at this scale, will recover it. Representing context evidently helps the model avoid the false alarms that pure lexical cues produce — a passage full of charged words that is not, in context, manipulative. It does not, at this scale, give the model the deeper competence required to recognise manipulation that hides in structure. The easy half of the contextual promise is delivered; the hard half is not.

6. Why Recall May Resist, and How Strong the Result Is

Several explanations are compatible with this outcome, and a single run cannot distinguish them. The first is capacity: DistilBERT is a small, distilled model, and a larger one (for instance RoBERTa in its larger configurations) might represent structural relations the distilled model cannot. The second is training budget: three epochs on CPU is a deliberately modest regime, and recall on the minority class is often the last thing to improve with more training. The third is truncation: capping inputs at 256 tokens removes the wider context in which some manipulation is only legible. The fourth is supervision: this model is trained on the binary label alone, discarding the natural-language explanations that PropXplain provides, which may be exactly the signal that distinguishes structural manipulation from ordinary charged language. The fifth is simple stochasticity: the F1 difference of 0.043, on 250 positive examples and from a single fixed-seed run, is modest enough that part of it may be noise. A responsible reading treats the precision gain as real and the recall result as a genuine non-improvement, while reserving strong claims about magnitude until the experiment is repeated across several seeds with confidence intervals.

7. Limitations

The result is a single evaluation run with a fixed seed and no confidence intervals; its magnitudes should not be over-read. The configuration is deliberately modest — a distilled model, few epochs, short sequences, CPU training — so it bounds what this particular transformer achieves here, not what transformers can achieve on the task. The labelling is binary and the corpus is drawn from a specific journalistic source under a specific guideline set, so neither the definition of 'propagandistic' nor the class distribution should be assumed to transfer. As before, the detector measures textual signatures and makes no claim about intent, coordination, or effect.

8. Conclusion

Replacing a lexical baseline with a fine-tuned transformer, on identical data and protocol, improves propaganda detection modestly and asymmetrically: it buys precision, not recall. Contextual representation teaches the detector to stop over-flagging, but it does not, at this scale, recover the subtle, structural manipulation whose existence motivated the move. The honest finding is thus a sharpened research question rather than a solved problem: the recovery of fine manipulation — raising recall on the hard cases without surrendering the precision now gained — is the open problem of the defensive audit layer of the Politomorphism Engine. The paths it points to are concrete: larger contextual models, explanation-level supervision from the corpus's own rationales, longer context windows, and repeated runs with confidence intervals. That this programme's second instrument reports a limit rather than a triumph is a feature of it, not a failure: a defensive instrument earns trust by stating precisely what it cannot yet see.

Reproducibility

The transformer results are produced by a single training-and-evaluation script (fine-tuning distilbert-base-uncased with a class-weighted objective, three epochs, learning rate 2e-5, maximum length 256, fixed seed), evaluated once on the held-out PropXplain English test partition with the same metric code as the baseline. The figures correspond to that one run. The complete code, pinned dependencies, and both working papers are maintained in the public repository at github.com/politomorphism/politomorphism-defense.

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